

Model Documentation and Audit Trail

Statistical Consulting ST4050

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1 Overview

This audit trail begins with a simple documentation of all of our files, and the structure in which we have organised them. We are working in the exact structure (including ordering) of files included in this GitHub: <https://github.com/mattmccarthy/Statistical-Consulting-/tree/main>. I would highly encourage you to open that alongside this document so you can see what we are referring to, and why we refer to it in this order.

We discuss files in the exact order they appear in the GitHub. This is solely to make it easier to follow alongside. It is not the order in which they were created, nor the order in which they should be ran (in most cases). We made this decision due to the sheer volume of code used throughout this project.

To improve auditability, our code is well structured, formatted incredibly repetitively, and commented heavily. Furthermore, each script utilises `set.seed(100)` to ensure reproducibility.

Our Repo consists of many types of files. This is due to the vast amount of comparison to be done with different objects. For every model we fit, we saved both the model object, and its test set so that we could compare objects. The reason we saved the test set in every case was so that we could ensure models were both being tested on the **same** unseen data, i.e. to confirm our `split_data` function was being utilised correctly.

Our Repo consists of 3 main folders:

- R, contains all scripts and objects related to modelling and creating plots etc,
- data, contains all data utilised, be it raw, processed or derived datasets, and
- figs, contains all figures utilised in our report.

Each of these folders consists of many subfolders, which is why we have formatted this report as sections being the main folders, subsections being subfolders, subsubsections being subsubfolders (this only happens in the case of Frequency and Severity Modelling) and individual files and objects being denoted as paragraph. This simply allows us to utilise the table of contents above as a quick navigation route around the file paths.

Each paragraph (individual file or object) consists of 3 categories:

- Type: The type of file we are discussing (e.g. RDS, CSV, R File).
- Importance: (out of 5) with regards to our final chosen model specifications.¹
- Description: a short overview of what the file does, and what we used it for.

Finally, only minor model documentation is included in this document. We documented our models extensively as we fit them, and included these write-ups in Section 4, 5, and 6 of our report. For extensive detail on individual model specifications and justification for all of the decisions we made, please refer to these sections. I can assure you that were we to include it all here once more, it would only serve to decrease auditability.

We hope you have more fun auditing our work than we had creating it.

¹Some models, such as decision trees were discarded early, and have no real bearing on the outcome of our report. They do not warrant as much attention as say our Negative Binomial GLM, or our Gamma IBLM.

2 R

2.1 Data Cleaning

`freq_data_cleaning`

Type: R Script

Importance: 4

Description: Contains all code we used to clean the `motor_policy_year_100k_2025-10-22` dataset. All cleaning decisions (e.g. imputation) are discussed in detail in Report Section 3.1.

`sev_data_cleaning`

Type: R Script

Importance: 4

Description: Contains all code used to clean the `motor_claims_100k_2025-10-22` dataset. All cleaning decisions (e.g. imputation) are discussed in detail in Report Section 3.1.

2.2 Fairness and Interpretability

`01_adjusting_for_fairness`

Type: R Script

Importance: 5

Description: Deals with the issue of gender being in the model. Addresses whether or not it is still being "included" through proxies. Backs up our assertion that our final models could reasonably satisfy real-world regulators provided more work was done.

`02_diagnostic_plots`

Type: R Script

Importance: 4

Description: Generates diagnostic plots in relation to exclusion of gender and proxies.

`03_additional_diagnostic_plots`

Type: R Script

Importance: 4

Description: Mimics the above, but adds extra detail we decided we should be discussing in the report. Each script does deal with slightly different metrics, so keeping separate made the most sense.

`04_interpretability_gamma_IBLM`

Type: R Script

Importance: 5

Description: Ensures that even our model including XGBoost is interpretable. Generates numerous key plots, such as SHAP for global dependence, local dependence, and even waterfall plots to show the explainability of our chosen model. Backs up our assertion that our ML model satisfied the end goal of the project. This is discussed thoroughly in Section 7.2 of our report.

2.3 Frequency Modelling

2.3.1 Comparison Scripts

negbin_vs_poissonIBLM_comparison_script

Type: R Script

Importance: 5

Description: Compares our Frequency Negative Binomial and Poisson IBLM models on key metrics, and motivates our decisions for proceeding with Negative Binomial.

2.3.2 Decision Trees

01_fit_freq_decision_trees

Type: R Script

Importance: 2

Description: Fits frequency decision trees using internal validation. Motivates our decision to not include decision trees as a model contender.

poisson_tree_train_pruned_1se_deep

Type: RDS File (Model Artefacts)

Importance: 2

Description: Deep pruned (using $1 - SE$) decision tree model object.

poisson_tree_train_pruned_1se_simple

Type: RDS File (Model Artefacts)

Importance: 2

Description: Shallow, interpretable pruned (using $1 - SE$) decision tree.

poisson_tree_train_unpruned_deep

Type: RDS File (Model Artefacts)

Importance: 2

Description: Deep unpruned decision tree model object.

poisson_tree_train_unpruned_simple

Type: RDS File (Model Artefacts)

Importance: 2

Description: Shallow, interpretable unpruned decision tree.

2.3.3 EDA

freq_age_eda

Type: R Script

Importance: 5

Description: Frequency EDA for policyholder age.

freq_area_eda

Type: R Script

Importance: 2

Description: Frequency EDA for Area.

freq_employment_missing_eda

Type: R Script

Importance: 3

Description: Frequency EDA for Employment Missing Binary flag. Used to conduct a MNAR (Missing Not at Random) investigation in Section 3 of report.

freq_fuel_eda

Type: R Script

Importance: 3

Description: Frequency EDA for vehicle fuel type. Motivates our decisions to retain EV category in our models.

freq_gender_eda

Type: R Script

Importance: 5

Description: Frequency EDA for policyholder gender.

freq_ncd_level_eda

Type: R Script

Importance: 4

Description: Frequency EDA for No-Claims discount level. Motivates some of our proxy investigation (Women claim less often, on average have higher NCD).

freq_num_drivers_eda

Type: R Script

Importance: 2

Description: Frequency EDA for number of drivers. Used to conduct an investigation into whether or not gaming was present in this dataset.

freq_occupation_eda

Type: R Script

Importance: 4

Description: Frequency EDA for policyholder occupation. Motivates our decision to aggregate to five risk buckets. No real loss in fit and requires 9 less predictors.

freq_overnight_parking_eda

Type: R Script

Importance: 1

Description: Frequency EDA for overnight parking location.

freq_province_eda

Type: R Script

Importance: 2

Description: Frequency EDA for province in which policyholder resides.

freq_security_device_eda

Type: R Script

Importance: 2

Description: Frequency EDA for whether or not a security device was present in a vehicle. Motivates our decision to exclude it from final models (No loss in fit through exclusion).

freq_vehicle_age_eda

Type: R Script

Importance: 4

Description: Frequency EDA for vehicle age. Motivates deeper investigation later on, specifically through the use of SHAP in Gamma IBLM interpretability section, allowing us to see non-linear effect our models predict.

freq_vehicle_power_eda

Type: R Script

Importance: 4

Description: Frequency EDA for vehicle power.

2.3.4 Hurdle

01_fit_validate_hurdle_model

Type: R Script

Importance: 2

Description: Fits and validates our frequency hurdle model. Motivates its exclusion as a final candidate model.

2.3.5 Lasso Regularisation

01_fit_validate_lasso

Type: R Script

Importance: 2

Description: Fits and validates our Lasso model. Motivates its exclusion as a final candidate model.

2.3.6 NegBin

01_negbin_fit

Type: R Script

Importance: 5

Description: Fits our initial Negative Binomial Model, before additional considerations such as gender exclusion. Deals with any and all issues that arose, e.g. multicollinearity.

02_negbin_test_agespline

Type: R Script

Importance: 4

Description: Initial tests we ran to determine the number of degrees of freedom to allow the age spline. Motivated our decision to decrease from 8 (initially supported by EDA) to 6, due to evidence of overfit.

03_negbin_validation1

Type: R Script

Importance: 5

Description: Validation for our Negative Binomial model, motivates final modelling decisions made throughout our report.

negbin_model

Type: RDS File (Model Artefacts)

Importance: 5

Description: Negative Binomial model used in pricing etc. Can be loaded into R to provide sample pricing calculations to demonstrate the efficacy of our models.

test_negbin

Type: RDS File (Dataset)

Importance: 5

Description: Test dataset for the Negative Binomial model. Saved for ease of loading into comparison scripts to ease comparability of models.

2.3.7 Poisson IBLM

01_fit_poisson_IBLM

Type: R Script

Importance: 5

Description: Fits our initial Poisson IBLM. Issues dealt with include lack of offset specification in the "IBLM" package. This script (discussed in extensive detail in our report) motivated our decision to exclude the Poisson IBLM from our final chosen models.

iblm__poisson__model

Type: RDS File (Model Artefacts)

Importance: 5

Description: Poisson IBLM model object. Can be loaded into R comparability scripts for ease of use. Large object, about 300Mb.

test_iblm

Type: RDS File (Dataset)

Importance: 5

Description: Test dataset for the Poisson IBLM, loaded into many comparability scripts to compare and contrast our specified models.

2.3.8 Quasi Poisson

01_fit_validate_quasi_poisson

Type: R Script

Importance: 3

Description: Fits and validates our Quasi-Poisson. Calculates low level metrics which informed our decision to exclude this from our final model selection.

2.4 Miscellaneous

Probability of Loss

Type: R Script

Importance: 1

Description: Important script to generate barplot for appendix.

Train-Test Validation Split

Type: R Script

Importance: 5

Description: 60/20/20 splitting function, including *set.seed(100)* implicitly. This function was used for splitting and training in almost every single model fit. This is how we ensured we retained the exact same test set every time we ran our models for ease of comparability.

XGBoost First Attempt

Type: R Script

Importance: 1

Description: First attempts at XGBoosting. Irrelevant to final conclusions of project.

2.5 Severity Modelling

2.5.1 Comparison Scripts

gammaGLM_vs_GammaIBLM_comparison_script

Type: R Script

Importance: 5

Description: Compares fit of Gamma GLM and IBLM on unseen (identical) test data. Minor pre-processing required as the IBLM took slightly different factor specifications due to how XGBoost handles factors. Motivated our decision to prefer the IBLM in our final pricing model.

2.5.2 DT

01_fit_decision_tree

Type: R Script

Importance: 3

Description: Fits a shallow, interpretable decision tree for claims severity. Metrics computed in this script lead to similar issues as those with the frequency model. The outcome is too noisy, decision tree only split on incredibly strong predictors such as age, leading to an unusable GI pricing model.

pruned_decision_tree

Type: RDS File (Model Artefacts)

Importance: 1

Description: Decision tree model object for loading into comparability scripts.

2.5.3 EDA

sev_area_eda

Type: R Script

Importance: 2

Description: Severity EDA for policyholder area. Motivates further desire for more granular data. Discussed in Limitations section of report.

sev_engine_cc_eda

Type: R Script

Importance: 3

Description: Severity EDA for Engine CC. Motivates our decision to not rely on it too heavily as a predictor (clear data issues, majority of data is 600cc).

sev_overnight_parking_eda

Type: R Script

Importance: 3

Description: Severity EDA for overnight parking location of vehicle.

sev_policyholder_age_eda

Type: R Script

Importance: 4

Description: Severity EDA for policyholder age. Motivates our continued spline specification of a natural spline on 4 degrees of freedom.

sev_security_device_eda

Type: R Script

Importance: 2

Description: Severity EDA for security device present in a vehicle. Motivates our decision to exclude from final pricing models (no material loss in fit through exclusion).

sev_usage_eda

Type: R Script

Importance: 4

Description: Severity EDA for vehicle usage. Motivated decision to include as an interaction term.

sev_vehicle_power_eda

Type: R Script

Importance: 4

Description: Severity EDA for vehicle power.

2.5.4 Gamma

01_fit_severity_gamma_glm

Type: R Script

Importance: 5

Description: Fits our Gamma GLM, before exclusion of immutable factors (gender).

02_error_metrics_gamma_glm

Type: R Script

Importance: 5

Description: Computes numerous error metrics which motivated us to include our Gamma GLM as one of our final contenders for our pricing model.

gamma_GLM

Type: RDS File (Model Artefacts)

Importance: 5

Description: Gamma GLM model object for easy loading into comparison scripts - eases comparability between distinct models.

test

Type: RDS File (Dataset)

Importance: 5

Description: Test dataset for the Gamma GLM model. Loaded into later comparability scripts for testing of distinct models.

2.5.5 Gamma IBLM

01_fit_gamma_IBLM

Type: R Script

Importance: 5

Description: Fits our specified Gamma IBLM, and performs XGBoost tuning over 3,488 different parametrisations, maintaining a low learning rate, and shallow depth, while also ensuring stable estimates. Selects the best fitting model, regardless of interpretability (this was done using the interpretability script below).

02_validation_gamma_IBLM

Type: R Script

Importance: 5

Description: Validates our chosen Gamma IBLM, ensuring that fit is sufficient, and all tuning was performed correctly.

03_interpretability_gamma_IBLM

Type: R Script

Importance: 5

Description: Investigates the interpretability of our Gamma IBLM, and generates a number of plots and tables discussed thoroughly in Section 7 of our report.

iblm_gamma_final

Type: RDS File (Model Artefacts)

Importance: 5

Description: Gamma IBLM model object for loading into comparability and pricing scripts. Running the model takes a substantial amount of computational expense, loading it in (despite its large file size) from GitHub was the optimal choice.

test_gamma_iblm

Type: RDS File (Dataset)

Importance: 5

Description: Test dataset for the Gamma IBLM. Used for loading into later comparability scripts.

train_gamma_iblm

Type: RDS File (Dataset)

Importance: 5

Description: Training dataset for the Gamma IBLM.

validation_gamma_iblm

Type: RDS File (Dataset)

Importance: 5

Description: Validation dataset for the Gamma IBLM.

2.5.6 XGBoost

01_fit_severity_xgboost

Type: R Script

Importance: 2

Description: Fitting script for our second XGBoost specification for Gamma Severity.

02_evaluate_severity_xgboost

Type: R Script

Importance: 2

Description: Evaluation script for our XGBoost severity model. Illustrated improvement in fit came at the cost of much interpretability and explainability, causing us to decide to move forward with the Gamma IBLM instead.

GBDT_GLM_Model_Comparison_V1

Type: R Script

Importance: 2

Description: XGBoost comparison script, to compare the Gamma GLM. Again illustrated this model was to be discarded, improvement in fit wasn't substantially better than Gamma, and was in fact worse in the tails, while losing much interpretability. Again further motivated us to proceed with other models.

gbdt_severity_xgb

Type: RDS File (Model Artefacts)

Importance: 1

Description: XGBoost model object to be loaded into separate scripts for ease of comparability.

3 data

3.1 derived

Contains derived datasets, i.e. datasets on which we performed feature engineering. For example, using k-means clustering to aggregate all occupations into 5 distinct risk buckets, reducing number of predictors whilst losing negligible material fit.

occupation_risk5_lookup

Type: CSV file

Importance: 5

Description: Lookup key for our occupation aggregation structure. Categorises each occupation into its respective risk bucket. Loaded into *policy_frequency_derived_script* to easily convert this factor.

policy_frequency_derived

Type: RDS File (Dataset)

Importance: 5

Description: Dataset used for final frequency modelling, utilising all feature engineering performed on the dataset.

policy_frequency_derived_script

Type: R Script

Importance: 5

Description: R code used to create the derived Frequency dataset. Ensures reproducibility and auditability of our method.

3.2 extras

ireland_province_claim_proportions

Type: CSV File

Importance: 2

Description: Proportion of policyholders in each province, split equally amongst their counties in order to create the heat maps used in the Geospatial Distribution of Claim section in our report. The lack of granular detail is discussed there, and further so in the Limitations section of our report.

3.3 processed

Contains processed datasets (i.e. data after Data Cleaning step). For each of these, we retained the CSV, and the RDS file to retain our desired factor specification up until this point in the initial modelling process.

claims

Type: CSV File

Importance: 4

Description: Cleaned *motor_claims_100k_2025-10-22* dataset, featuring all feature engineering and data cleaning up until the conclusion of initial modelling section of report.

claims_severity

Type: RDS File

Importance: 5

Description: As above, except formatted as an RDS to both save memory, and retain factor specifications.

policy

Type: CSV File

Importance: 4

Description: Cleaned *motor_policy_year_100k_2025-10-22* dataset, featuring all feature engineering and data cleaning up until the conclusion of initial modelling section of report.

policy_frequency

Type: RDS File

Importance: 5

Description: As above, except formatted as an RDS to both save memory, and retain factor specifications.

3.4 raw

motor_claims_100k_2025-10-22

Type: CSV File

Importance: 5

Description: Raw severity dataset provided by the SAI. Both grouped and ungrouped data dictionaries outlining this dataset are included in the Appendix of our main report.

motor_policy_year_100k_2025-10-22

Type: CSV File

Importance: 5

Description: Raw frequency dataset provided by the SAI. Both grouped and ungrouped data dictionaries outlining this dataset are included in the Appendix of our main report.

README_initial_2025-10-08

Type: Markdown File

Importance: 5

Description: Initial README file supplied by the SAI, outlining the structure of the data.

4 **figs**

This folder contains each plot we designed in R to illustrate our points more effectively in our report. For both of our sanity, I will not list them all individually here. The direct link to the folder is: <https://github.com/mattmccarthy/Statistical-Consulting-/tree/main/figs>, should you wish to browse our plots independent of their report sections.

Due to our repeated use of *set.seed(100)*, all of these plots are easily reproducible to validate our outcomes.

5 Conclusion

We hope that through a combination of this audit trail, and our extensive model documentation included in our final report, that you can get a great sense of our decisions and methods utilised throughout this project with relative ease.

Should you have any outstanding questions, you may contact each of us here:

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